

Andrew Thomas

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Personal Statement

A final-year student at Lancaster University studying Electronic Engineering. Experienced in working with STM, Arduino and Texas Instruments microcontrollers, and applications involving communication protocols such as I2C, SPI, and CAN.

Interested in exploring AI applications using PyTorch. I have created convolutional neural networks for image categorization using datasets from Kaggle, such as the apple leaf dataset. Proficient in algorithmic problem-solving using python on Leetcode.

I am part of the UKESF scholarship scheme which has helped me take part in many STEM outreach activities. Always seeking new challenges and opportunities to further develop my skills and knowledge in the field of electronic engineering as I embark on my career.

Skills

- C Embedded Programming
- Python PySimpleGUI, Pytorch, Numpy and Pandas
- PCB Analog & Digital Design (Altium&KiCad)
- Laser Milling and Engraving
- Soldering, Sintering and Wire-bonding
- Advanced-numerical skills (MATLAB)
- Clean Room Experience
- PCB Milling and Fabrication
- Ansys and Comsol Thermal-mechanical Simulation
- UV-Vis and FTIR spectroscopy
- TFT Probing and Analysis

Experience

September 2022 – Ongoing

Lancaster Formula Student, Lancaster University – Instrumentation Lead

- Currently taking a lead role in the Lancaster E-Racing team to design a new telemetry to integrate seamlessly with the EV electrical systems and improve understanding of vehicle performance. This involves the design of a new CAN-based, Sensor and datalogging network, which has not previously been included in the vehicle. Various electrical and mechanical characteristics and vehicle diagnostics will be logged on race-day.
- Working on a redesign and optimization of previous teams LV circuits. Supporting other team members in their tasks such as full LV circuit testing and testbench vehicle start-up while collecting telemetry data from motor and battery controllers.

July 2021 – July 2022

Compound Semiconductor Applications (CSA) Catapult, Newport – Advanced Packaging Intern

- Internship at CSA Catapult working on research with industrial partners to develop the UK compound semiconductor capabilities in sectors such as EV, Telecoms, space and healthcare. Developed skills in automatic and manual wire bonding while applying process development to micro-scale MIMIC and VCSEL die. This involved optimising bonding and loop parameters to meet the bonding pad compliance and short project deadlines.
- Designed a Python GUI to determine Power efficiency of RF Amplifier devices by integrating RF analysers and generators and DC Power Supplies using PyVISA; which uses the VISA interface.
- Developed thermo-mechanical simulation models to study experimental device stresses and optimise packaging design. Designed a high-power gate driver with embedded die packaging technologies. Used state of the art processes such as silver sintering and materials such as copper-diamond to solve the problems involved with the ever-increasing power densities of smaller SiC MOSFET and Si IGBT dies.
- Operated in all the major CSA technology discipline labs: packaging, photonics, RF and Power. Worked with industrial partners to develop new compound semi-conductor research and applications and develop innovative prototype devices.

March 2020 – March 2021

Running and Athletics Club, Lancaster University – Training Captain

- Elected Training Captain for the Lancaster University Running and Athletics Club. Created a training plan for each training group containing members of similar ability to ensure club members could get the most out of the high intensity sessions. This was planned to ensure that our top athletes peaked during the correct time in the season.

September 2019 – March 2020

Morecambe Foodbank, Lancaster University – Foodbank Volunteer

- Had a leading role in Foodbank collections on campus as well as the subsequent delivery of the food to Morecambe. I organised the collection of boxes at each college.
- Permission for door-to-door collection as well as using college space for food boxes required formal permission from colleges weeks in advance. Friendliness and sincerity were essential to make the most out of the collection.

September 2020 – October 2020

Refugee Community Kitchen, Calais – Volunteer

- Involved cooking for large groups of people (300+) on a strict timeline. Working with a team to create quality meals from the limited resources provided and communicating clearly to ensure efficient delivery of the service.
- We delivered and served the food in refugee camps across the Calais and Dunkirk area with a special consideration for dignity, respect and professionalism.

Education

SEPTEMBER 2018 – DECEMBER 2022

Lancaster University - Electrical and Electronic Engineering with 2 Scholarships

Part 1 – 69.2 %

Modules include: Engineering Management, Electrical Circuits and Power Systems, Electromagnetics & RF Engineering, Digital Electronics, Power Electronics and Applications, Integrated Circuit Engineering, Optoelectronics and wireless communications & Digital Signal Processing.

Part 2 – 65.8 %

Modules include: Advanced Embedded Systems, Terahertz Science and Technology, Microengineering & High frequency electronics.

SEPTEMBER 2012 – JULY 2018

Epsom College, Epsom

A-Levels in Maths, Further Maths, Chemistry and Physics (**3 A* & 1 A**)

JULY 2008 – JULY 2012

Downsend School, Leatherhead

GCSEs in Art, Biology, Chemistry, Chinese, English Language, English Literature, French, Geography, Mathematics, Physics (**2 A* , 3 A , 4 B & 1 C**)

Personal Interests

I am very interested in keeping up with research in physics, cosmology and Machine learning. Channels on YouTube such as Sabine Hossenfelder and PBS Space Time help cultivate my interests in scientific research. I enjoy reading classic fiction and ancient history and philosophy. I like to play classical piano music in my spare time. My particularly favourite composers are Sergei Bortkiewicz and Benjamin Godard. I enjoy running and taking part in local running events such as Parkrun.